

Jiahui LI

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PROFILE

A highly motivated researcher with a strong foundation in Artificial Intelligence and Software Engineering, specialising in reinforcement learning and LLMs with demonstrated expertise in LLM complex reasoning within intricate label-free settings. First author of a ICLR 2027 submission, with a proven track record of independent research and resolving complex technical challenges.

Research Interests: Artificial Intelligence, Machine Learning, Transfer Learning, Agents, Embodied AI

EDUCATIONAL BACKGROUND

Sun Yat-sen University

09/2023 - 06/2027

Bachelor of Engineering in Software Engineering

GPA: 4.0/4.0 | **Average Mark:** 94/100 | **Rank** 2%/231

Relevant Modules: Linear Algebra (99), Digital Image Processing (99), Principles of Database Systems (98), Principles of Compilers (97), Computer Graphics (97), Algorithm Design and Analysis (96), Machine Learning and Data Mining (95)

PAPERS

Forthcoming Article: Li, J., et al. (Under Review). Where to Verify, Where to Explore: Test-time reinforcement learning with confidence-conditioned verification. *ICLR 2027*.

- **Objective:** To address the under-explored yet critical optimisation of Pass@k measures in a completely label-free setting. This is achieved by developing a novel confidence-adaptive framework that expands Pass@k coverage, mitigates pseudo-label noise, and significantly enhances the complex reasoning abilities of large language models.
- **Responsibilities:** Led the conceptualisation of the confidence-conditioned mechanism based on the key insight that verification capability generally leads generation capability, adaptively allocating verification resources across high, medium, and low-confidence samples. Furthermore, designed an exploration-enhancing length-diversity reward to prevent trajectory collapse in high-confidence samples, and prepared the complete manuscript as a co-first author for submission to *ICLR 2027*.

RESEARCH EXPERIENCE

Academic Research Project: Optimisation of Test-Time Reinforcement Learning in Label-Free Scenarios

Principal Investigator

10/2025 - Present

- Built upon the open-source distributed reinforcement learning framework (verl) to develop the TTRL-CoCoV architecture, successfully implementing the principles of confidence-conditioned verification into a fully functional code base.
- Conducted a comprehensive set of ablation experiments on six widely-recognised reasoning benchmarks (e.g., MATH500, AIME24/25) to identify and confirm the individual contributions of the length-diversity reward and noise-filtering components.
- Diagnosed and resolved training instabilities during prolonged metric collapse, iteratively refining the algorithmic design and experimental paradigm, which resulted in an average absolute gain of +18.7% in Pass@16 and +9.8% in Pass@1 compared to the baseline method.

Academic Research Project: Bi-Level Optimisation Framework for Privacy-Preserving Fine-Tuning in Medical Large Models

Principal Investigator

06/2025 - 08/2025

- Developed a bi-level optimisation architecture featuring a hyperparameter search at the upper level and LoRA-based multi-task fine-tuning at the lower level, explicitly targeting privacy leakage reduction while preserving task performance.
- Created a proxy loss function based on the evaluation metrics of privacy attacks, which dynamically modified parameters using a regularised search in order to create an iterative loop of “fine-tune, evaluate, and re-tune” parameters.
- Successfully tested the framework under harsh testing conditions, reducing privacy leakage indicators by 83.5% while improving the win ratio by 61.9%.

National College Students' Innovation Entrepreneurial Training Plan Programme: Uncertainty-Aware Data Generation Methods for Large Model Preference Alignment

Core Member

01/2025 - 11/2025

- Designed an uncertainty-aware adaptive screening algorithm to mitigate the unwarranted occurrence of pseudo-label noise during the mathematical reasoning processes conducted by large language model (LLM) systems.
- Developed a stability measurement for reasoning based on Top-K local entropy standard deviation and Z-score cross-difficulty normalisation, enabling dynamic screening and weighted voting of paths.
- Employed extensive correlation analyses and ablation experiments to validate efficacy of the algorithms, producing a conclusive report resulting in the National Award for Outstanding Conclusion.

AWARDS & HONOURS

Scholarships & Honours:

- National Scholarship, 12/2025
- National Scholarship, 12/2024
- First-Class Scholarship, Sun Yat-sen University, 12/2025
- Outstanding League Member, Sun Yat-sen University, 07/2025
- First-Class Scholarship, Sun Yat-sen University, 12/2024

Prizes:

- Outstanding Conclusion, National College Students' Innovation Entrepreneurial Training Plan Programme, 2025

Awarded for contributions as a Core Member on the Uncertainty-Aware Data Generation Methods for Large Model Preference Alignment.

- Provincial First Prize, Contemporary Undergraduate Mathematical Contest in Modelling, 2024
- Provincial Second Prize, The Chinese Mathematics Competitions for College Students, 2024
- Silver Award, Sun Yat-sen University Programming Contest, 2024
- Bronze Award, Sun Yat-sen University Programming Contest, 2023
- Excellence Award, 2nd SecretFlow Cup, 2025

Awarded for contributions as a Principal Investigator on the Bi-Level Optimisation Framework for Privacy-Preserving Fine-Tuning in Medical Large Models.

SKILLS

- **Languages:** Mandarin (Native), English (Proficient).
- **Programming Languages:** Proficient in Python and C++, with extensive experience in version control (Git) and modern development environments (VS Code).
- **Artificial Intelligence Frameworks:** Adept at utilising distributed reinforcement learning frameworks (e.g., verl, PyTorch), with comprehensive expertise in the end-to-end fine-tuning of large language models and the implementation of reinforcement learning algorithms.
- **Academic Competencies:** Proficient in comprehending advanced academic literature in English; possesses strong analytical capabilities for conducting rigorous hypothesis-driven research, correlation analysis, and resolving complex algorithmic bottlenecks.